A Dual-Tier Edge-Cloud Architecture for Real-Time Basketball Activity Recognition and Fatigue Monitoring

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***Abstract*—Traditional sports wearables rely predominantly on kinematic data from Inertial Measurement Units (IMUs), lacking the physiological context required to classify complex movements and detect muscle fatigue. This paper presents a multi-modal wearable ecosystem designed for basketball analytics, fusing 9-axis IMU data with Electromyography (EMG) via a bilateral footwear prototype. The system utilizes a dual-tier architecture: a quantized TinyML neural network executing locally on an Android edge-computing hub for real-time inference, and a cloud-based Random Forest classifier for deep post-session analysis. Empirical evaluation demonstrates that EMG features, specifically Variance (VAR) and Root Mean Square (RMS), are the primary predictors for movement classification, outperforming traditional kinematics. During live testing, the localized TinyML model achieved 79.5% accuracy, outperforming the cloud-tier model while eliminating network latency. Furthermore, a 500ms majority-voting smoothing algorithm increased classification stability by 6.4%, providing a robust framework for real-time athletic monitoring and fatigue-based injury prevention.**

***Index Terms*—Wearables, Sensor Fusion, EMG, IMU, Edge ML, TinyML, Basketball Analytics.**

I. Introduction

The constant pursuit of efficiency and durability extends not only to technology but also to human performance. Central to this evolution is the rapid progression of wearable technology, which has transitioned from simple step-counting peripherals to sophisticated diagnostic tools. These devices now encompass a vast range of applications, including medical diagnostics, injury prevention, sleep analysis, and high-fidelity sports performance monitoring.

In recent years, the synergy between advanced microcontrollers and smartphone has revolutionized the sports landscape. Modern athletics is increasingly defined by data-driven decision-making, where statistics are utilized to gain competitive advantages. However, as the demand for deeper insights grows, the complexity of the data required to satisfy these metrics has increased proportionally.

Compared to single-activity recognition, team sports present significant challenges for real-time analytics. The inherent unpredictability, high-intensity intervals, and constant physical interaction between players create a volatile data environment. These sports are characterized by explosive, short-duration movements and high levels of physiological stress, especially during a game's climax where the risk of injury is most prevalent [1].

Despite these advancements, standard sports wearables remain limited by their reliance on single-modality sensing, typically utilizing only Inertial Measurement Units (IMU). While kinematics provide a trajectory of movement, they lack the neuromuscular context required to differentiate between explosive intent and mere mechanical motion. Furthermore, the reliance on cloud-based processing introduces latency that can be critical in injury-prevention scenarios.

This paper addresses these gaps by proposing a multi-modal, dual-tier architecture that fuses Inertial Measurement Units' (IMU) kinematics with Electromyography (EMG). By leveraging Edge Artificial Intelligence for localized inference on a mobile hub, the system provides a low-latency solution for movement classification and fatigue monitoring, ensuring that the physiological cost of movement is captured as accurately as its physical path.

II. Related Work

The development of intelligent sports wearables has increasingly focused on objective physiological monitoring to supplement traditional kinematic tracking. Recent research demonstrates a paradigm shift toward multi-modal sensor fusion and edge-based analytics to enhance the accuracy of activity recognition and fatigue detection.

A common challenge in athletic monitoring is the inability of single-modality sensors to provide full biomechanical context. Hwang et al. addressed this by integrating surface electromyography (sEMG) and inertial measurement units (IMU) to classify physical fatigue during natural over-ground walking. Their hybrid Convolutional Neural Networks-Long Short-Term Memory (CNN-LSTM)-Attention model, evaluated via leave-one-subject-out cross-validation, achieved 87.94% accuracy, proving that combining physiological markers from the gastrocnemius with kinematic jerk signals significantly enhances system robustness against individual variability [2]. Similarly, Otálora et al. demonstrated that for upper limb fatigue estimation during repetitive lifting, a combination

of IMUs and Optical Fiber Sensors (OFS) achieved 95.4% accuracy. Their findings highlight that while EMG is a gold standard for bio-electrical muscle function, its reliability in long-term work environments can be compromised by factors like skin perspiration, making supplemental kinematic data essential [3].

The choice of classifier and processing pipeline is critical for real-time mobile deployment. Baigarayeva et al. developed an IoT-based system, "Qimyl," which performs on-device feature extraction (RMS, MAV, median frequency (MDF)) to classify muscle fatigue during sustained isometric tasks. By comparing logistic regression, decision trees (DT), and XGBoost, they identified DT as achieving the highest accuracy (90.7%) for their specific submaximal grip tasks. Furthermore, their work emphasizes the value of synchronizing objective EMG features with subjective self-assessments (e.g., the Borg scale) to improve the ecological validity of the models [4].

Beyond standard classification, researchers have explored deterministic indices to objectively detect the moment of failure. Qassim et al. proposed a double-step binary classifier utilizing a new fatigue index based on the difference between high-frequency and low-frequency components of the sEMG signal. This algorithm achieved 94.66% accuracy in identifying the moment a muscle fails to sustain required force, providing a blueprint for active injury-prevention alerts [5]. These findings are supported by Kamaruddin et al., whose review of surface electromyography emphasizes that while time-domain features like Waveform Length (WL) offer low computational complexity, frequency-domain analysis remains superior for quantifying the physiological shift to lower frequencies that characterizes neuromuscular fatigue [6].

III. System Architecture

The proposed system follows a decentralized, multi-modal architecture designed to bridge localized physiological sensing with high-level cloud analytics. The framework is divided into three primary layers: the sensor-embedded hardware, the mobile edge-computing hub, and the cloud-based inference server highlighted in yellow, blue and green respectfully in Fig. 1.

A. Overview

As illustrated in Fig. 1, the architecture facilitates a high-bandwidth data pipeline that transitions from raw physical signals to actionable biofeedback. The process is categorized into three distinct operational stages:

1) **Acquisition Layer:** The footwear hardware captures bilateral IMU and EMG signals, calculates six EMG features, and sends the data to the next layer through BLE.
2) **Edge Computing Hub:** The Android application serves as the connection point between the user, the prototypes and the cloud. It handles the simultaneous asynchronous streams from both devices, performing temporal synchronization to align the 15 features from each foot into a unified 30-feature vector. This is where the cramping warning and the localized TinyML inference occur, allowing for "offline" movement recognition and immediate feedback to the athlete.
3) **Cloud and Analytical Layer:** The mobile app forwards the synchronized data to a cloud infrastructure. Data is stored in a time-series database (InfluxDB) for historical visualization, while the Flask-based server executes the high-dimensional Random Forest model.

B. Hardware Prototype

The sensory layer consists of two footwear-mounted prototypes. Each unit is centered around an Arduino Nano 33 BLE Sense Rev2, chosen for its embedded 9-axis Inertial Measurement Unit (IMU) and Bluetooth Low Energy (BLE) module (Fig. 2). To provide physiological context, a localized Electromyography (EMG) sensor is integrated into each module to monitor the electrical activity of the Gastrocnemius Lateralis.

The firmware implements a non-blocking sampling strategy. To maintain sampling periodicity despite the periodic blocking of the BLE radio stack, a microsecond-precision hardware timer tracks "time debt." If a transmission delays the loop, a sequential catch-up routine is executed to preserve the 1ms EMG sampling interval.

To overcome the bandwidth limitations associated with transmitting raw high-frequency raw EMG data, six EMG features are sent instead. Every 20 ms, the system samples the IMU data and computes the six time-domain features using a 200ms sample window (N=200):

Root Mean Square (RMS): Represents the signal power and is defined as:

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2} \tag{1}$$

Variance (VAR): Measures the power of the AC component of the EMG signal:

$$VAR = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu)^2 \tag{2}$$

Mean Absolute Value (MAV): An indicator of muscle contraction levels.

$$MAV = \frac{1}{N}\sum_{i=1}^{N}|x_i| \tag{3}$$

Waveform Length (WL): Captures the cumulative complexity of the signal's amplitude, frequency, and duration.

$$WL = \sum_{i=1}^{N-1}|x_{i+1} - x_i| \tag{4}$$

Zero Crossing (ZC) and Slope Sign Change (SSC): These provide a simple representation of the frequency domain properties in the time domain, helping to identify shifts in muscle firing rates.

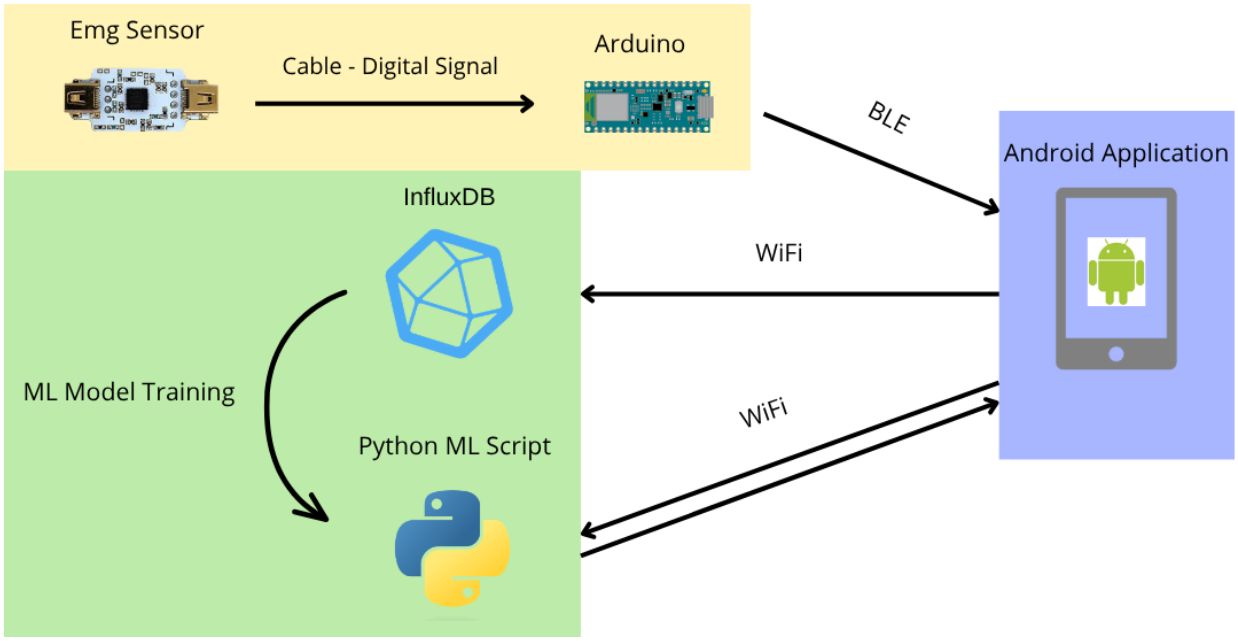

Fig. 1. System Overview.

$$ZC = \left\{ \sum_{n=2}^{N} sgn(x_i \cdot x_{i-1})(x_i - x_{i-1}) \geq \epsilon \right. \tag{5}$$

$$sgn(x) = \begin{cases} 1 & \text{if } x > \epsilon \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

$$SSC = \sum_{i=2}^{N-1} f[(x_i - x_{i-1}) \cdot (x_i - x_{i+1})] \tag{7}$$

To optimize BLE bandwidth, the system bundles five 30-byte frames into a single 150-byte payload, transmitted to the mobile hub every 100ms.

C. Android Application and Data Synchronization

The center of the proposed architecture is a native Android application developed in Kotlin using the Model-View-ViewModel (MVVM) architecture. It serves as the bridge between the user and the prototypes, handling the bilateral data synchronization. When the app receives the BLE communications from the left or right device, it separates the five received data bundles and assigns time-stamps to each proceeding to forward them to the cloud InfluxDB data base.

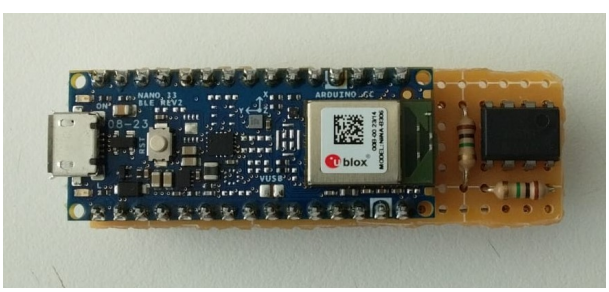

Fig. 2. Prototype Picture.

Beyond data processing, the application manages individualized user profiles that may ensure personalized monitoring in future iterations and the sensors' calibration. To maintain high performance during intense activity, the application utilizes asynchronous background threads for inference and cloud synchronization, ensuring the real-time activity dashboard remains responsive.

D. Temporal Smoothing and Output Stability

To handle the inherent mechanical noise of high-intensity basketball movements, the system utilizes a 300ms sliding window with a 100ms step size for feature aggregation. A critical contribution of this architecture is the implementation of a post-processing temporal smoothing layer. Utilizing a continuous sliding window (implemented via an *ArrayDeque*), a Majority Voting algorithm evaluates the last five local predictions, providing a 500ms context. This logic acts as a digital low-pass filter, effectively eliminating prediction "flickering" and hardware-induced false positives, thereby ensuring a stable and reliable stream of activity classifications during steady-state movements.

E. Dual-Tier Machine Learning

The architecture employs a dual-tier approach, for movement recognition, to compare latency and model complexity:

- Edge Tier (TinyML): A lightweight, quantized Neural Network (NN) is embedded directly within the Android application using TensorFlow Lite. This model provides ultra-low latency inference, enabling real-time feedback and driving the deterministic injury-prevention algorithm independently of network availability.

- Cloud Tier (Random Forest): Simultaneously, the 300ms window is serialized into a JSON payload and transmitted via an asynchronous HTTP POST request to a Flask-based server. The server hosts a high-dimensional Random Forest (RF) classifier used for deep analysis and comparative validation against the edge model.

F. Fatigue Monitoring and Warning

Beyond activity classification, the system implements a real-time monitoring layer designed to identify early indicators of neuromuscular fatigue. This functionality is supported by the observation that localized muscle fatigue manifests as a measurable increase in the amplitude of the EMG signal, due to the need of higher muscle effort to maintain force output.

The monitoring logic operates on a threshold-based trigger system integrated within the Android app. This threshold is personal and a calibration should be performed in the beginning of every session. For this paper, only one player was tested so only one threshold was calculated and the calibration wasn't implemented. The threshold value was calculated by monitoring the athlete at the beginning of practice (with relaxed muscles) and when he was almost cramping and comparing the data while idling or walking.

To prevent false positives caused by mechanical spikes or explosive movements (such as a single rebound), the alert logic requires a temporal persistence of high-intensity sEMG activity. A warning is only triggered if the physiological features exceed the pre-set threshold for a continuous duration of 1 second. Once these conditions are met, the application provides immediate feedback with a visual alert on the dashboard, enabling the athlete or coach to make informed decisions regarding rest and injury prevention.

IV. Experimental Method and Results

A. Experimental Method

The system was evaluated through a series of dynamic basketball drills involving several volunteer subjects. The dataset was partitioned into an 80/20 train-test split to validate the classification models across five macro-movements: Idle, Walking, Running, Defensive Slide, and Rebound. The cramping prevention system was only tested on one player that pushed his muscles to near involuntary contractions.

The prototypes were mounted directly in the athletes' footwear to approximate the ideal positioning (on the bottom of the feet as an insole or a sock). The devices were secured using the shoe laces and the powerbank was secured in the users socks, utilizing the existing infrastructure. With this configuration, the IMU accurately captures the kinematics of the foot, such as accelerations and rotational velocities during the athletes' movements. Furthermore, this setup minimizes the mechanical strain exerted by the power cables on the USB ports, while keeping the EMG leads in close proximity to the chosen muscle group.

The muscle chosen to be measured was the Gastrocnemius Lateralis as it is one of the most important muscles for explosive jumping motion. The position for the EMG sensors

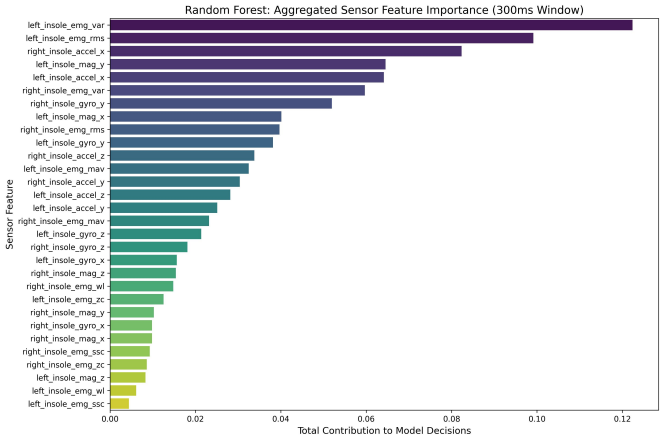

Fig. 3. Feature Importance.

was chosen as recommended by the European Union (EU)'s SENIAM (Surface ElectroMyoGraphy for the Non-Invasive Assessment of Muscles) project ([7]) ensuring standardization and maximization of the signal-to-noise ratio.

To ensure correct labeling for the supervised learning models, a manual annotation interface was integrated into the mobile application. Data collection was performed in controlled sessions where the athlete executed specific basketball movements while researcher chose the correct label and turned a "Send data" switch within the Android User Interface (UI).

When the switch was activated, the application bundled the incoming 30-feature sensor vector with the corresponding label (e.g., Defensive Sliding or Running), user ID and time, before transmitting the packet to the InfluxDB database.

B. Model Training and Convergence

The TinyML model demonstrated robust learning, achieving a validation accuracy of approximately 94high generalization. Conversely, the Cloud-based Random Forest (RF) was evaluated using Receiver-Operating Characteristic (ROC) and Precision-Recall (PR) curves. The Area Under the Curve (AUC) for all classes approached 1.0, and the PR curves confirmed the model's ability to handle the class imbalance inherent in basketball data (e.g., rare rebounding events vs. frequent running/walking).

C. Feature Importance and Sensor Contribution

A core contribution of this work is the validation of EMG integration. Gini Feature Importance analysis (Fig. 3) revealed that EMG-derived features, specifically VAR and RMS, were the primary contributors to classification accuracy. This confirms that physiological electrical activity provides critical intent-based context that kinematics alone lack. Additionally, the model exhibited a preference for features from the dominant (left) limb, successfully capturing subject-specific biomechanical asymmetries.

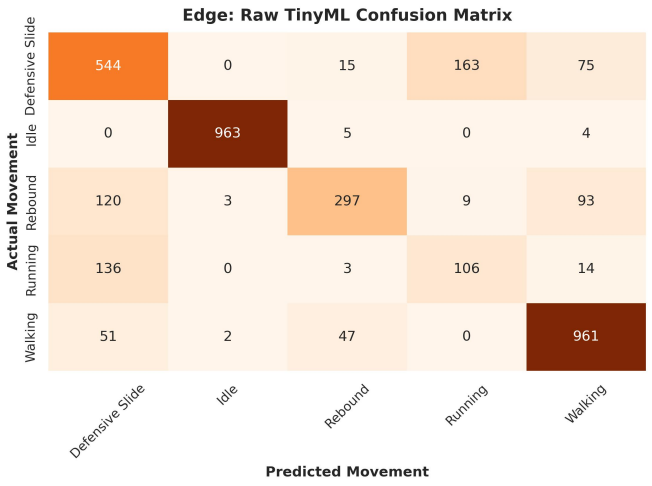

Fig. 4. TinyML Confusion Matrix.

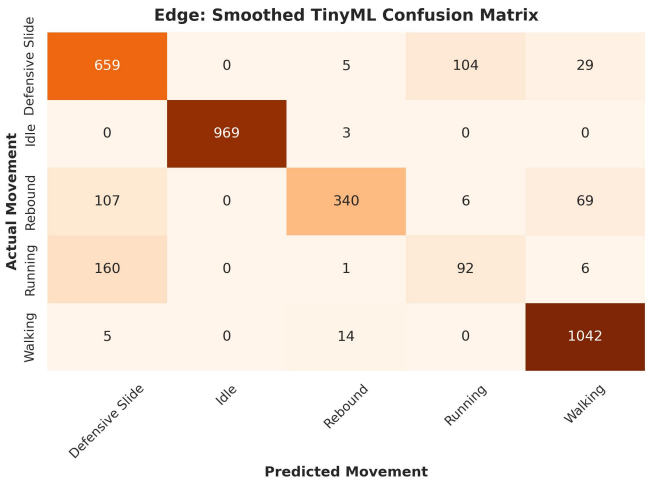

Fig. 5. TinyML with Smoothing Confusion Matrix.

D. Edge vs. Cloud: Real-Time Performance

During live testing, the localized TinyML model outperformed the Cloud RF model, achieving an overall accuracy of 79.5% compared to the server's 71.2%. While the RF model excelled at slower movements (Idle, Walking), it struggled with dynamic transitions in the field. Notably, the TinyML model provided instantaneous inference, whereas the RF model introduced measurable network latency.

E. Temporal Smoothing Improvements

The 500ms majority-voting algorithm provided a significant stability boost. For the TinyML model, smoothing increased overall accuracy by 6.4% by acting as a digital low-pass filter that neutralized prediction flickering. Confusion matrix analysis (Fig. 4, Fig. 5) shows that while smoothing improved steady-state detection, it reduced sensitivity to transient events like "Rebounding," which were often outvoted by surrounding states. This identifies a fundamental trade-off: smoothing is essential for consistent activity tracking but may require dynamic windowing for explosive event detection in future iterations.

V. Discussion and Know Limitations

The experimental results validate the core hypothesis that integrating EMG with IMU data significantly enhances movement recognition. Feature Importance analysis confirms that RF classifier prioritized the features VAR and RMS over kinematic data to resolve biomechanical similarities between movements.

Furthermore, the 500ms Majority Voting window proved to be an effective digital low-pass filter, increasing "Idling" accuracy by 6.4%. However, this introduces a localized "responsiveness-stability" trade-off. While the resulting latency is acceptable for general activity tracking and fatigue monitoring, the suppression of explosive movements (e.g., rebounds) suggests that future iterations should employ adaptive windowing to maintain sensitivity.

The transition to live basketball testing exposed critical durability issues. The mechanical stress exerted by the wires ripped the USB port of one prototype, necessitating manual hardware reinforcement. Furthermore, the prototype occasionally shifted, introducing mechanical noise. Data integrity for the left limb was specifically impacted by a degradation in the BLE antenna range following hardware strain, leading to periodic packet loss when the android device was not in immediate proximity. Ergonomically, the external power management circuit caused minor discomfort, indicating that a transition to fully integrated, soft-shell enclosures is necessary for long-term athletic use.

The current dataset faces generalizability constraints due to the limited sample size. The models are likely tuned to the specific biomechanical signatures of the test subjects, requiring a broader demographic pool for universal validation. Additionally, the manual labeling process introduced "boundary noise" during movement transitions, slightly affecting training data purity. Finally, the physiological capture of pre-cramp states proved difficult to reproduce safely and consistently. This resulted in challenges for establishing a sustained RMS baseline, making the fatigue monitoring alerts highly sensitive to individual calibration.

VI. Conclusion

In summary, the localized TinyML model outperformed the Cloud-tier RF (79.5% vs 71.2% accuracy), while providing the instantaneous feedback required for injury prevention. The implementation of a 500ms majority-voting algorithm successfully improved classifications despite the identified trade-off in sensitivity regarding explosive movements like rebounds.

Future work will focus on transitioning the prototype into a fully integrated smart-textile garment to improve ergonomics and signal stability. Additionally, expanding the dataset demographic and correlating EMG threshold breaches with clinical physiological markers will further refine the system's predictive accuracy and universal applicability.

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